Comparison between different machine learning api libraries

Features	H2O	SparkMLlib	Sparkling Water	Weka
Open source/closed source	Open	Open	Open	Open, but licensed under GPL 2.0 for commercial use
Algorithms available	Deep learning, generalised linear modelling, gradient boosting machine, K-Means, standard statistics like min, max, mean, stddev	Classification, regression, clustering, collaborative filtering, dimensionality reduction, optimisation primitives	Deep learning, generalised linear modelling, gradient boosting machine, K-Means, standard statistics like min, max, mean, stddev	Classifiers, clusterers, associations, attribute selection, preprocessing filters
Languages supported	R, Phtyon, Scala, Java	Python, Scala, Java	R, Python, Scala, Java	Java, Octave/Matlab, R, Python
Efficiency with large datasets	Moderate	Good	Good	Moderate
Scalable	Yes	Yes	Yes	No
Community support	Good	Good	Good	Good

To implement these steps, many application programming interface (API) libraries are available. The table above shows a comparison between H2O, Spark MLlib, Sparkling Water, Weka platforms and API libraries. H2O is supported by Python, R, Scala and Java programming languages. SparkMLlib library is available in Python, Scala and Java languages.

H2O is a fully open source, distributed in-memory ML platform with linear scalability. It supports popular ML algorithms including deep learning and gradient boosted machines. H2O also has AutoML functionality that automatically runs through all algorithms and their hyper parameters to produce correct logical answers from models. More than 18,000 companies are using this platform. R and Python communities are heavily using H2O platform.

Spark is a fast and general-processing engine compatible with Hadoop data. It can run in Hadoop clusters through Spark standalone. Spark can process data in Hadoop File System (HDFS), HBase, Hive, Cassandra and other Hadoop input formats. MLlib is a special API library in Spark engine. It supports classification, logistic regression, Naïve Bayes, Random Forest, K-Means and Gaussian mixture ML algorithms. This library can be used in Python or Scala programming language.

To know whether ML can be applied for the Internet of Things (IoT) processing, we have to understand abnormal situations that can occur in IoT networks. On the application layer of the IoT, broken sensors might send wrong sensed values to a connected server continuously. If the intelligent network is not aware of this problem, it will not be able to detect accidents in IoT networks.

ML can be applied on sensor networks. ISM band is used by 802.11, 802.15.4 and 802.15.1 wireless protocols. When different devices are scattered in one area, there is a different system interference. ML can be used to classify wireless channel errors in different categories, and this can be used to diagnose problems in IoT networks.

In wireless sensor networks, fault detection is a complicated process. Faults caused by degraded or malfunctioning sensors are classified as data faults. Fault types caused by low battery, calibration, communication or connection failures are classified as system faults. When ML is applied, faults such as gain, out-of-bound, offset and stuck-at can be recognised in detail. Classifying faults can also help in IoT networks.

It is not easy to design fault-tolerant systems in constrained IoT networks. People are working on designing IoT systems that can detect, predict and recover from abnormal situations. ML is helping improve IoT networks, too. END

By: Tularam M. Bansod

The author is proprietor of TMB Enterprises, a startup focused on embedded systems. He has authored two books on microcontroller programming.

This article was originally published in the July 2019 issue of Electronics For You.